SessionSignal

Predicting which online shopping visits end in a purchase — DES423 group project

> **We predict whether someone shopping on a website will actually buy something — just by watching how they browse.**

What goes in

We look at **one visit** to an online store and ask simple questions:

- How many product pages did they look at?
- How long did they stay on the site?
- Did they leave immediately, or keep browsing?
- Are they a new visitor or a returning one?
- What month is it?

We never look at what is in their cart or how much money they spent. Only their browsing behaviour.

What comes out

A single percentage. *"This visitor has a 64% chance of buying."* That is it — one number per visit.

Why anyone would want this

An online store gets thousands of visits a day, and **about 85% of them leave without buying anything.**

The store's analyst wants to work out *why* people leave — but nobody can review 10,000 visits by hand. They have time for maybe twenty.

So which twenty? Right now they guess. Our model sorts the visits so the interesting ones float to the top.

Does it work?

44%	**16%**	**2.79×**
of the visits our model ranks highest are real buyers	if the analyst just picked visits at random	better use of the same hour of work

The best part

Sometimes the model says *"this person will probably buy"* — and they don't. That looks like a mistake. **It isn't.** That is exactly the visit worth investigating.

> **A real example from our data.** Someone looked at **38 product pages** and spent **over an hour** on the site. They never bounced once. Our model gave them a 64% chance of buying.
>
> **They bought nothing.**

Something went wrong for that person. Maybe shipping cost too much. Maybe the checkout was confusing. We don't know *why* — but we found them inside a pile of 12,000 visits, and now a human can go and look.

How we built it

The data

12,330 real shopping visits from a public dataset published by the UCI Machine Learning Repository. Each row is one visit; 15.5% of them ended in a purchase.

Honest testing

We held back **2,441 visits** that the model never saw during training, and only scored them once at the very end. That is why the numbers are trustworthy.

Comparing models

We did not rely on one method. We trained **four** and compared them on exactly the same test data.

The winner was a **random forest** — hundreds of small decision trees that each make a guess and then vote together.

MODEL WE TRIED	WHAT IT DOES	SCORE
Always say “no purchase”	A dummy baseline, to prove the others are worth it	0.156
Logistic regression	Draws one straight dividing line through the data	0.331
Random forest	Many decision trees vote together	0.372
Gradient boosting	Trees added one by one, each fixing the last one's errors	0.370
Random forest, tuned ← we used this	**Same as above, with settings optimised**	**0.374**

The score is **average precision**. We deliberately avoid “accuracy”: a lazy model that says “nobody will buy” is **84% accurate** and completely useless, because it never finds a single buyer.

What we are careful NOT to claim

- ✗ We don't know *why* a shopper left
- ✗ We don't promise the store will earn more money
- ✗ It can't run live while someone is still shopping — only after they leave
- ✓ We find the visits worth investigating
- ✓ We prove it works on data the model never saw
- ✓ A human still does the actual investigating

If a judge asks “what is your project?”

“Most people who visit an online store leave without buying. Stores can't review every lost visit, so we built a model that scores each visit on how close it came to a purchase. It lets an analyst spend their time on the visits that actually mattered — we found it surfaces almost three times more real buyers than checking at random.”

What every variable means

One row of the dataset is one visit. These are the 17 inputs plus the answer.

VARIABLE	IN PLAIN WORDS	RANGE IN OUR DATA
PAGES THEY OPENED, AND HOW LONG THEY STAYED		
Administrative	Account and admin pages opened — login, sign-up, contact us, shipping info	0–27 (median 1)
Administrative_Duration	Seconds spent on those pages	0–3,399 s
Informational	Info pages — About Us, FAQ, help, store policy	0–24 (median 0)
Informational_Duration	Seconds spent on those pages	0–2,549 s
ProductRelated	Actual product pages viewed — the strongest shopping signal	0–705 (median 18)
ProductRelated_Duration	Seconds spent browsing products	0–63,974 s
QUALITY OF THE VISIT, MEASURED BY THE STORE'S ANALYTICS		
BounceRates	Average bounce rate of the pages seen. A “bounce” is landing on a page and leaving without clicking anything. Higher is worse	0–0.20
ExitRates	Average exit rate of the pages seen — of everyone who viewed that page, the share who left the site from it. Higher is worse	0–0.20
PageValues	**Removed on purpose.** Average revenue value of pages visited before a purchase. It is built from revenue, so keeping it would let the model peek at the answer	0–361.76
WHEN THEY VISITED		
SpecialDay	How close the visit was to a shopping holiday such as Valentine’s or Mother’s Day. 0 is a normal day, 1 is right on it	six values 0–1; 90% are 0
Month	Month of the visit	10 months (no Jan or Apr)
Weekend	Was it a Saturday or Sunday	23.3% were weekends
WHO THEY WERE — THE FOUR ANONYMISED CODES		
OperatingSystems	Which operating system, **but the real names were removed by the publishers.** Code 2 is the most common (6,601 visits) so it is probably Windows — that is a guess, not a fact	codes 1–8
Browser	Which browser, anonymised the same way. Code 2 dominates with 7,961 visits	codes 1–13
Region	Which geographic region of the country. We are not told which is which	codes 1–9
TrafficType	How they arrived — search, direct, advert, email. Also anonymised	codes 1–20
VisitorType	Returning visitor (85.6%, buys 13.9%), new visitor (13.7%, buys **24.9%**), or **Other** (0.7%) — an analytics catch-all for visitors the tracker could not classify, usually blocked or cleared cookies. The dataset never defines it	3 categories
WHAT WE PREDICT		
Revenue	**The answer. True means this visit ended in a purchase**	**15.5% are True**

If someone asks what "Operating System 3" is: we do not know, and neither does anyone using this dataset — the publishers anonymised it. The model can still use it, because what matters is that visitors on code 3 behave differently from code 2, not what code 3 is called.

The team

STUDENT ID	NAME	MAIN CONTRIBUTION
6622782298	Sirawich Narakulsookpipat	Problem research and framing
6622780342	Chaitawat Boonkitticharoen	Dataset audit and target definition
6622782215	Bhuchi Kumprakob	Exploratory data analysis
6622780276	Nonthaphat Chuathong	Evaluation design and model selection
6622770657	Kamin Sirisuwapong	PyCaret comparison and hyperparameter tuning
6622772455	Poonnawit Wongprateepwilai	Results analysis and business solution
6622790267	Choltawat Jullakasevi	Streamlit app, demonstration and limitations

Sources Dataset: Sakar, C. & Kastro, Y. (2018). *Online Shoppers Purchasing Intention Dataset*, UCI Machine Learning Repository (DOI 10.24432/C5F88Q), CC BY 4.0. Market size: eMarketer, global retail ecommerce $6.42 trillion in 2025, 20.5% of all retail. Cart abandonment: Baymard Institute, 70.22% unweighted average of 50 studies published 2006–2025. All model metrics come from `artifacts/results.json` and are reproducible by running `train.py`.